WASTEWISE SCAN · USER REQUIREMENTS

WasteWise Scan user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First user requirements (kit 1.8 upgrade), drawn from the users and stakeholders in WSC-PRB-001 v0.5 Tables 1 and 2

WasteWise Scan user requirements

What users need, written from their side. Product requirements that meet these needs are in WSC-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix WSC-RTM-001 (`03b-traceability.md`). The needs come from the users and stakeholders in WSC-PRB-001 Tables 1 and 2. None has yet been confirmed in a co-design session with the first candidate partner (a member-owned waste picker cooperative, first candidate SWaCH in Pune, decided by Amish on 2026-10-02), so every priority is a proposal. Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Right resin	When I scan a plastic item I am told whether it is PET, HDPE, PP, PS or PVC, and it is right almost every time it gives an answer	High	Waste picker; cooperative sorter; small recycler or buyer (PRB Table 1)
UR-002	Performance	Quick	The answer comes in about a second, so checking doubtful items does not slow work paid by weight	High	Waste picker; cooperative sorter (PRB Table 1)
UR-003	Performance	Honest unknown	When it cannot tell, for black, wet or very thin items, it says "unknown" rather than guessing, and it is never used to judge whether waste is safe to handle	High	Small recycler or buyer; cooperative (PRB Table 1, Out of scope)
UR-004	Environment and durability	Works in the sun	It works outdoors in strong sun at streets, dumpsites and yards	High	Waste picker (PRB Table 1 context)
UR-005	Logistics	Lasts a shift	It lasts a full shift with no power at the work site and charges from a phone charger	High	Waste picker; cooperative (PRB constraints)
UR-006	Usability	One hand, light	I can hold it in one hand, press one button and carry it all shift	High	Waste picker (PRB constraints)
UR-007	Usability	No reading needed	I can read the result at arm's length in sun without reading text: colour, icon and resin code, in my own language	High	Waste picker; trainer (PRB constraints, literacy)
UR-008	Usability	Our price grade	It shows the grade our buyer pays for, from a price table the cooperative sets and edits, offline	High	Cooperative; small recycler or buyer (PRB Table 1)
UR-009	Performance	A record of scans	The cooperative can keep and export a record of what was scanned, for training and for disputes	Medium	Cooperative; trainer, NGO or researcher (PRB Table 1)
UR-010	Environment and durability	Survives handling	It survives drops, dust, rain and heat in daily use	High	Waste picker; cooperative (PRB constraints)
UR-011	Maintenance	Easy to keep right	A quick self-check keeps it accurate, and it tells me when a check is due	Medium	Cooperative sorter (PRB Table 1)
UR-012	Cost	Affordable to a cooperative	A cooperative can afford a shared scanner (USD 150 value-engineering target, decided 2026-10-02)	High	Cooperative; NGOs (PRB constraints)
UR-013	Performance	Works with the phone app	Items it cannot resolve can be passed to WasteWise-ml on a phone, and the two records join	Medium	Cooperative; WasteWise-ml (PRB prior work, Stakeholders)
UR-014	Safety	Safe invisible light	The invisible infrared light cannot harm my eyes or anyone near me	High	Waste picker; municipality (PRB Table 2)

ID	Category	Title	Need	Priority	Source
UR-015	Safety	Safe battery	The battery is safe to carry, drop and charge in a hot climate	High	Waste picker; cooperative (PRB constraints)
UR-016	Deployment and assembly	Built locally	A local maker or makerspace can build and repair it from off-the-shelf modules and printed parts	Medium	Trainer, NGO or researcher; Helpful Engineering makers (PRB constraints, Stakeholders)
UR-017	Regulatory	Our data, our consent	Scan records and spectra stay on the device unless we agree to share them, on terms our organization co-owns	Medium	Cooperative; waste pickers (WSC-DDR-002, decided 2026-10-02)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.